Gabriel Paredes Mendoza

SOFTWARE DEVELOPMENT | ICT | MECHATRONICS

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PROFESSIONAL PROFILE

Multidisciplinary professional with experience in software development, mechatronics, manufacturing and hospitality. Developed web and mobile applications, ERP modules, security-log platforms and IoT systems. Combines technical delivery with production operations, customer service, community training and academic work in ICT analysis and telecommunications.

TECHNICAL SKILLS

Languages: Dart, Python, JavaScript, TypeScript, PHP, C/C++.

Applications: Flutter, React, React Native, Flask, Django, Node.js, Odoo, Vite, Expo, Expo Application Services, Skia.

Data & infrastructure: PostgreSQL, MySQL, Firebase, Elasticsearch, Logstash, Kibana, Filebeat, Docker, Kubernetes, AWS, Terraform, CI/CD.

Engineering: Arduino, microcontrollers, PLCs, SolidWorks, Fusion 360, CAD/CAM.

ICT analysis: Requirements elicitation, stakeholder engagement, risk assessment, network design.

Languages spoken: Spanish (native), English (B2).

PROFESSIONAL EXPERIENCE - SOFTWARE DEVELOPMENT

Full Stack Developer

Duodata | Ecuador / remote from Australia | Dec 2021 - Jan 2025 | Full-time

Custom ERP modules - Ecuadorian Navy

Technologies / methods: Odoo, Python, PostgreSQL

- Developed customised ERP modules and integrated them with existing systems to support resource allocation, operational workflows and data exchange.

Security log data warehouse

Technologies / methods: Docker, Python, Elasticsearch, Logstash, Kibana, Filebeat, MinIO

- Built ingestion pipelines for Sophos, FortiGate, Cisco and Postfix logs. Developed Python reporting scripts and Kibana plugins for PDF reports, dashboard migration and firewall administration. Configured event notifications and Elasticsearch backups; upgraded the stack from 8.4.2 to 8.14.2.

ZKTeco biometric administration platform

Technologies / methods: Flutter, Python, Flask, Node.js, MySQL, Docker

- Developed a cross-platform administration application for user management, event logs and remote door control. Implemented device-protocol communication and three microservices for REST APIs, command generation and database synchronisation. Added user-photo transfer, Flutter unit tests and CI/CD pipelines. Increased throughput by 40% in operations processed per minute and reduced end-to-end system response time by 30%. Achieved 90% Flutter test coverage, reduced errors detected during testing by 35%, and cut deployment time by 50% through CI/CD.

PROFESSIONAL EXPERIENCE - SOFTWARE DEVELOPMENT

Full Stack Developer (continued)

Duodata | Ecuador / remote from Australia | Dec 2021 - Jan 2025 | Full-time

The Hit - music social application

Technologies / methods: React Native, Expo, Redux-Saga, ESLint

- Contributed to a mobile social network for music sharing and artist chat. Implemented asynchronous operations and contact-based relationship metadata to support user discovery, collaborating with developers and designers. Contributed to an application with over 30,000 combined Android and iOS downloads; improved asynchronous-operation handling time by 15-25% and contact-based search results by 10%.

Frecuento - mobile commerce application

Technologies / methods: React Native, Expo, Redux-Saga, ESLint

- Developed features for product browsing and credit-card purchasing on Android and iOS. Applied MVC architecture, asynchronous state management, testing and linting to improve code organisation and maintenance. Contributed to an application with over 10,000 Android downloads and 15,000 iOS downloads. Reduced controller/view code by 30-40% through MVC and asynchronous/state-management code by 20-30% using Redux-Saga.

Founder & Full Stack Developer

GP-Developer | Ecuador | Feb 2020 - Dec 2021 | Own business

- Independently developed approximately 15 technology projects for clients.

GP-Security - IoT vehicle security platform

Technologies / methods: Flutter, Firebase, AWS, Terraform, Docker, Python, Flask, TCP/IP, Arduino, C/C++, Fusion 360

- Designed an end-to-end vehicle tracking platform connecting custom GPS hardware, cloud services and a cross-platform application. Implemented location history, geofencing, remote engine immobilisation, window control and automated security alerts.
- Built Python REST and multithreaded TCP services, improving processing speed by 35% and reducing latency by 25%. Implemented CanTrack protocol communication for custom and third-party GPS devices.
- Developed Flutter applications for web, Android and iOS. Achieved 95% unit/widget test coverage and reduced post-release defects by 40%; automated delivery with CI/CD.
- Designed a Fusion 360 enclosure, assembled electronic circuits and programmed embedded functionality using ATmega, SIMCom and Espressif platforms.

Full Stack Developer

CODENORTE | Ecuador | Jan 2019 - Feb 2020 | Full-time

IoT GPS tracking system

Technologies / methods: PHP, MySQL, HTML, ATmega, SIMCom

- Developed a web interface and database for GPS data. Programmed tracking hardware to collect and transmit coordinates, integrating embedded devices, backend services and data storage.

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PROFESSIONAL CV

ACADEMIC RESEARCH - MECHATRONICS

Mechatronics Research Collaborator

Intelligent Systems Research Group, Universidad Técnica del Norte | Ecuador | Jan 2017 - Feb 2019 | Academic collaboration

Passive exoskeleton development

Technologies / methods: LabVIEW, National Instruments controllers, PCB design, CAD/CAM

- Contributed to PCB design, mechanical design and manufacturing. Maintained control software, developed system functionality and reworked signal acquisition for National Instruments controllers; provided technical troubleshooting.

Three-axis CNC drawing plotter

Technologies / methods: SolidWorks, Arduino, C/C++, GRBL, G-code

- Designed and assembled a drawing plotter, integrating mechanical structures, stepper motors, drivers and control electronics. Implemented coordinated motion from vector designs and calibrated and tested the prototype.

Automated production line integration

Technologies / methods: Siemens PLC, SMC FMS-200, pneumatics, sensors

- Assembled modular manufacturing stations and integrated electrical wiring, sensors, pneumatic actuators and PLC controllers. Supported commissioning, calibration, troubleshooting and coordinated station testing.

Tablet counting and bottle filling prototype

Technologies / methods: Arduino, C/C++, infrared and photoelectric sensors

- Designed the conveyor and vibratory hopper, assembled electronics and programmed bottle detection, tablet counting and configurable filling cycles. Calibrated sensors and tested mechanical, electronic and software integration.

Custom mechanical parts and CNC manufacturing

Technologies / methods: SolidWorks, CAD/CAM, G-code, CNC milling and turning, 3D printing

- Reverse-engineered components through dimensional measurement. Created 3D models and technical drawings, prepared machining instructions and manufactured parts using CNC machines and additive manufacturing.

EDUCATION

Advanced Diploma of Information Technology (Telecommunications Network Engineering)

ICT60220 | National Institute of Technology, Australia | Completed 2025

Diploma of Information Technology (Cybersecurity)

ICT50220 | National Institute of Technology, Australia | Completed 2024

Degree in Mechatronics Engineering

Universidad Técnica del Norte, Ecuador | 2014 - 2019 | Graduated 2019

ACADEMIC PROJECTS - ICT ANALYSIS AND NETWORKS

National Institute of Technology (NIT) | Australia

Jul 2024 - Jun 2026 | Academic project work

Critical infrastructure cybersecurity requirements

Technologies / methods: BABOK v3, ISO/IEC 27001 principles, Essential Eight

- Defined scope, governance and stakeholder responsibilities. Elicited security and compliance requirements and prepared a prioritised baseline, traceability approach, risk register and change-control process.

Website accessibility redevelopment

Technologies / methods: WCAG, requirements management, Australian legislation

- Analysed accessibility issues, user needs and technical constraints. Documented functional and non-functional requirements, acceptance criteria, stakeholder roles and approval processes.

ICT privacy, intellectual property and ethics review

Technologies / methods: Policy analysis, ACS Code of Ethics

- Reviewed data handling, software licensing, copyright and ethical conduct. Updated governance procedures and prepared approval, communication and implementation materials.

LAN-Alpha telecommunications requirements

Technologies / methods: RSTP, VLANs, VoIP, DNS/DHCP, AAA, WHS

- Captured requirements for a mining-site network, including underground connectivity, real-time communications, redundancy and security. Documented scope, constraints, components and compliance considerations.

Link-Beta high-availability transmission link

Technologies / methods: Optical fibre, transmission architecture, change control

- Analysed a 100 km link between Perth NOC and Boonanarring Mine Site. Documented a 99.99% availability requirement, capacity and resilience needs, technology alternatives and delivery constraints.

Multi-site IP addressing and VLAN design

Technologies / methods: IPv4, IPv6, VLSM, VLANs, WAN addressing

- Produced addressing plans for six VLANs and three WAN links across Subiaco and Applecross. Calculated subnet capacity and documented network IDs, host ranges, masks and traffic segmentation.

DMC transmission network architecture

Technologies / methods: Cambium LINKPlanner, QGIS, optical and RF communications

- Prepared preliminary designs for Perth HQ, Gosnells Quarry and Cataby Mine Site. Evaluated fibre and radio options, selected equipment and performed link-budget analysis.

Cybersecurity resilience and continuity

Technologies / methods: Risk assessment, incident response, disaster recovery

- Analysed cyber risks and recovery needs. Prepared business continuity, disaster recovery and incident-response documents, with responsibilities, communications and evaluation materials.

BrightConnect digital transformation proposal

Technologies / methods: SWOT, stakeholder analysis, business case development

- Assessed learning delivery, student records, connectivity and administration. Proposed a cloud LMS, digital records, improved Wi-Fi and AI-supported administration, with risk and feasibility analysis.

ICT strategic gap and change assessment

Technologies / methods: Gap analysis, change assessment, action planning

- Compared ICT capability with organisational strategy. Identified gaps, impacts and resource needs, and prepared action plans and stakeholder communications to support prioritisation.

PROFESSIONAL EXPERIENCE - PRODUCTION AND HOSPITALITY

Fencing Production Operator

AJ Fencing Supplies | Australia | Oct 2023 - Feb 2026 | Casual

- Operated roll-forming lines for steel fence sheets, rails and posts; entered cut lengths and quantities and adjusted production controls.
- Loaded steel coils onto decoilers and moved materials with a counterbalance forklift. Cut posts and rails with a cold saw, inspected products and prepared dispatch pallets.
- Followed WHS procedures, lifting practices and PPE requirements; maintained a safe, organised production area.

Bartender

The Knock Restaurant | Australia | Feb 2025 - Present | Casual

- Prepared classic and contemporary cocktails and personalised recipes. Coordinated restaurant service, recommended beverages and maintained an organised bar.

Bartender

Mr Basil Rockingham | Australia | 2024 | Casual

- Prepared cocktails and customised recipes, managed concurrent orders during busy service and maintained beverage quality and attentive customer service.

Bartender

Encore | Australia | 2024 | Casual

- Prepared cocktails and beverages, managed orders efficiently and supported guests with recommendations while maintaining a welcoming bar.

Bartender

MiniBar | Australia | 2024 | Casual

- Served cocktails and spirits, provided personalised recommendations and coordinated with colleagues to support efficient service.

Bartender

Martini & Co | Australia | 2023 - 2024 | Casual

- Prepared classic and contemporary cocktails and maintained beverage presentation standards in a fast-paced hospitality environment.

Bartender

Santos y Pecadores | Australia | 2023 - 2024 | Casual

- Prepared cocktails and mixed drinks, supported beverage selection and maintained inventory and cleanliness during high-volume service.

Bartender

MIXING Bar | Ecuador | 2019 - 2023 | Shift and event work

- Created seven signature cocktails adopted into the official drinks menu. Experimented with flavour and presentation, provided personalised service and maintained beverage standards.

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PROFESSIONAL CV

LEADERSHIP AND VOLUNTEERING

NIT Connect Ambassador / Volunteer

National Institute of Technology | Australia | Ambassador: 2025 - 2026 | Volunteer: 2024 - 2025

- Personally delivered 20 digital literacy and cybersecurity awareness workshops for a total of 249 participants.
- Coordinated student volunteers and developed learning resources. Liaised with Perth libraries and community organisations; mentored students to deliver sessions independently.

Leadership, outreach and membership

IEEE | Ecuador / Latin America & Caribbean / global outreach | 2015 - 2021

Roles and recognition

Vice Chair, IEEEXtreme Public Relations Committee (2019 - 2021)
Lead Ambassador, IEEEXtreme Latin America & Caribbean (2018 - 2019)
IEEE MadC Ambassador (2018 - 2019)
Outstanding Member, IEEE Ecuador Section (2017 - 2018)
Lead Ambassador, IEEEXtreme Ecuador Section (2017 - 2018)
President, IEEE RAS UTN Chapter (2016 - 2017)
IEEE Member (2015 - 2021)

- Supported international public relations and outreach for IEEEXtreme, coordinating communication to promote the programming competition.

Robotics and Programming Volunteer Mentor

Oviedo High School | Ibarra, Ecuador | 2013 - 2014

- Taught robotics and programming and prepared students for a national competition, where the team achieved second place.

TRAINING AND CREDENTIALS

Forklift Licence (LF) | WorkSafe WA | 2025 | Current

Responsible Service of Alcohol (RSA) | EOT | 2023

FoodSafe Food Handler Certificate | Environmental Health Australia | 2023

Basic, Advanced and Web Flutter | Udemy | 2019

Human Talent Assistant | COORED | 2018

Website Development | C-DAC | 2017

Siemens and Industrial Networks | AiControl | 2017

CCNA Routing & Switching | Cisco | Certified 2016 | Not renewed

Administration with Linux | C-DAC | 2016

Java Development | C-DAC | 2016

Ethical Hacking and Penetration Testing | Green Ethics | 2016

ISO Programming for CNC | UTN | 2016

REFERENCES

Available on request.